the second and Rs 500 for the third, before the job will be confirmed and the ‘trainee’ tag will be sent to the crushing mill. I said yes.

I needed not more than Rs 100 to meet my expenses, and sent the rest home. It wasn’t a job I wanted to do, but it wasn’t bad in the beginning. I had to oversee everything from cutting and crushing of sugarcane, till the juice was turned into sugar crystals through a series of simple processes. Sugarcane came from the fields, fresh—the sugar yield can be as high as 10% of the cane weight if they reach the factory within a couple of hours; the longer it took, the poorer the yield. From the yard where truckloads arrived, sugarcane was moved on a rail-like system to the mill where it was shredded and crushed before the juice was sent to an evaporator, pan and a centrifuge where sugar crystallized. Bagasse and molasses were the by-products. While molasses went to distillers, the sugar factory used bagasse to fire a boiler that generated steam which in turn went into the power system that ran the plant.

A stationery maker, Seshasayee Paper Boards, found better use for bagasse—to make a fine variety of paper—and entered into a deal with the sugar factory. The paper mill would take bagasse from the sugar factory, and install an oil-fired boiler to produce steam for the factory. Witnessing this barter was to contribute much to my striking a deal with the French, several years later, for the development of the liquid propulsion Vikas engine which remains at the core of every successful PSLV launch by the Indian Space Research Organisation (ISRO).

The technology used at the sugar factory was so simple that I learned every bit of the operation in a couple of weeks. Anyone with a basic knowledge of fluid dynamics, heat transfer, steam engines and applied mechanics could master the working of a sugar factory easily. Work became less challenging and more boring, but I laboured on. To keep myself occupied, I explored all nooks of the workplace, including the smelly bagasse pit where engineers feared to tread. And there came a turning point of my job.

Karuppan, a man so true to his name dark, was in the pit when I sauntered in well ahead of my duty time. Karuppan was strenuously digging